

Functional& non-functional requirements:

Functional Requirements (User and System REQs) :

1	User shall be able to login using his fingerprint 1.1 user will enter his fingerprint 1.2 if the fingerprint is correct, system will work and display services 1.3 if the fingerprint is wrong, system will display error message
2	User should enter PIN code to do transactions 2.1 user should enter PIN code which consists of eight digits to do further transactions 2.2 The ATM authenticates the user with a pin code before allowing transactions. 2.3 If the PIN is invalid, the ATM system will display an error message 2.4 If the PIN is valid, the request will be transferred to the next service
3	User can have more than one account 3.1 user can have saving account, current account and gold account
4	User shall be able to withdraw cash 4.1 user will enter the amount of the withdrawal

	4.2 user will mention from which account he wants to withdraw (saving account, current account and gold account) 4.3 If the user's amount is less than the account balance, system will display success message and dispense the withdraw amount to the user then The new balance is forwarded to printer to print out the receipt 4.4 if the amount entered by the user is greater than the account balance, an error message is displayed
5	User shall be able to deposit 5.1 user will enter the amount of deposit . 5.2 system will display an success message 5.3 the new balance is forwarded to printer to print out the receipt 5.4 If the user wishes to make another transaction, the main menu is displayed. 5.5 If the user doesn't want to make another transaction, the system exits.
6	User shall be able to transfer money to various accounts 6.1 user will enter the account number that he wants to transfer to him 6.2 user will enter the amount of transfer 6.3 If the receiver's account number is valid and the user's amount is less than his account balance, system will display success message and

	The new balance is forwarded to printer to print out the receipt 6.4 if the amount entered by the user is greater than the account balance or the receiver's account invalid, an error message is displayed
7	User shall be able to view balance 7.1 user will mention from which account he wants to view the balance (saving account, current account and gold account) 7.2 The balance is forwarded to printer to print out the receipt. 7.3 The receipt contain this data (account id, account name, account type, the balance)
8	User shall be able to view the 5 last transactions 8.1 The 5 last transactions is forwarded to printer to print out the receipt 8.2 The receipt contain this data (account id, account name, account type, transaction type, data of transaction, time of transaction balance before transaction, balance after transactions)
9	System can display messages to the user at the end of transaction 9.1 System displays a success message to the user if the transaction complete successfully 9.2 system displays Failure message to the user If transaction failed under any reason

10	The system shall be able to send The transaction details to printer to print out the receipt 10.1 the receipt contain basic information like (Account id, account name, account type, transaction type, data of transaction, time of transaction)
11	The system should be integrated with bank API To enroll transactions
12	The admin shall be able to startup system 12.1 The admin will startup system at 7:00 AM by click on startup button
13	The admin shall be able to shutdown system 13.1 The admin will shutdown system at 11:00 PM by click on shutdown button
14	The admin can Refill ATM balance 14.1 the admin will check the balance of the ATM 14.2 if the balance less than 10000\$ the admin will Refill balance with minimum amount 500000\$
15	The system should be able to block user 15.1 system will block user after three times of login attempts failure
16	The system should be able to use user's behavior to check for unusual patterns 7.1 system should calculate the user's characteristics based on his spending pattern and country 7.2 system should store patterns for each user

	7.3 based upon stored patterns, system can recognize unusual patterns 7.4 If a deviation is detected in patterns (let's say more than 20-30), system will transfer user to security page
17	The system should require the user to provide additional verification 16.1 The system require the user to provide additional verification if unusual pattern is detected 16.2 user will transferred to security page 16.3 in the security page user will provide additional information like (national ID, phone, account password) 16.4 if additional information mapped the user, The user returned to his transaction 16.5 if additional information not mapped the user, system will cancel transaction and block user

Non-Functional Requirements :

Performance requirement [BR]

1.	each request shall be processed within 3 seconds
2.	The product shall, on average, operate without failure for 30 days.
3.	The restart time after failure should not exceed 0.5 minutes

Usability Requirements [AR]

4.	The system shall be easy to use on the first attempt by a member of the public without training
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5.	It should be possible to use the system to pay in different currencies.
6.	The number of times the user presses the support button must not exceed three times during one hour of use.

Safety and security requirements [SSR]

7.	The system shall distinguish between authorized and non-authorized
8.	system should recognize unusual patterns to verify user information
9.	The database should be secured from SQL injection to prevent leak or loss information
10.	After three times of login attempts failure , a security system may lock an account to protect a user's information from potential hackers.
11.	The system shall ensure that only authorized users are able to gain access.
12.	users Can just read information but they can't edit or Modify Anything except their personal information.
13.	The system shall be able to protect users against threats and attacks

Reliability Requirements [RR]

14.	The system services shall be available when it's requested by the users.
15.	The system should be provide catching of exceptions so that unintended results do not occur such system crashes or data validation failures.

Cultural REQs.

16.	The language used in the interface should be formal and polite
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Maintainability & Support REQs:

17.	The system shall be used on any ATM
18.	The system shall be able to be modified to cope with changes to central bank laws

Look-and-feel REQs.

19.	The system shall use only two colors.
20.	The system shall be attractive to users.

Legal REQs.

21.	The system shall Implement user privacy
22.	The system shall be complied with the central bank laws